

subtitle Basic idea about DC to AC converter
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subject Power Electronics PC-EE504

Inverter Lecture 02

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Lecture 02

- ① Brief recap of last lecture
- ② 1-ph full bridge VSI inverter
- ③ simulation study
- ④ comparative study between half and full bridge
- ⑤ References

1-ph full bridge VSI inverter

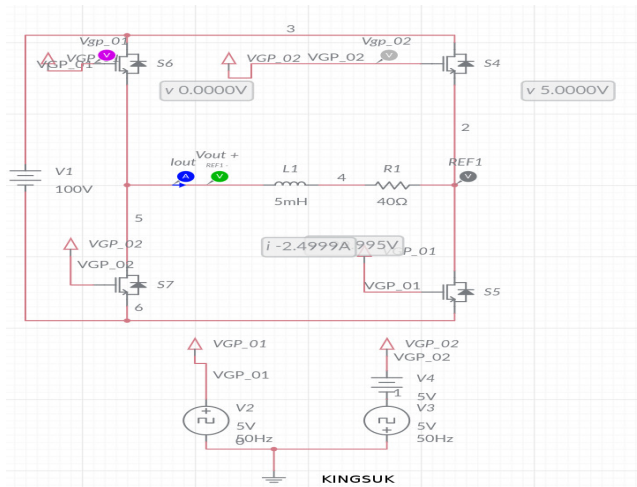


Figure: Full Bridge 1-ph Inverter

cont...

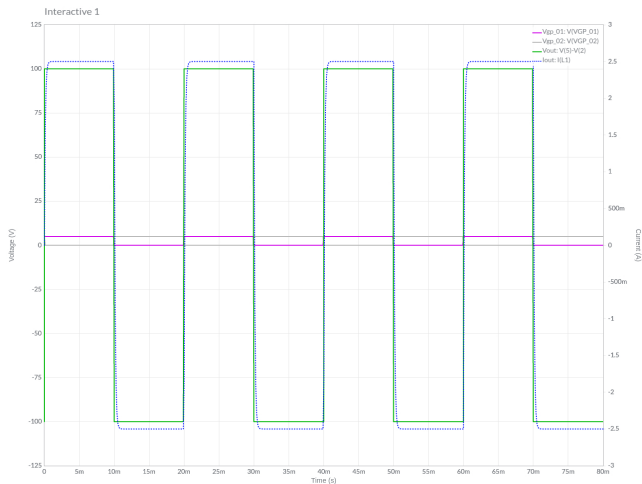


Figure: Voltage waveform

Mathematics of VSI full bridge

$$v_0 = \begin{cases} V_s; & 0 < t < \frac{T_s}{2} \\ -V_s; & \frac{T_s}{2} < t < T_s \end{cases} \quad (1)$$

$$V_{0ave} = V_s \quad (2)$$

cont...

Again, the instantaneous voltage can be written as

$$v_0 = \sum_{h=1}^{\infty} \frac{4V_s}{h\pi} \sin h\omega t \quad (3)$$

where $h=1,2,3\dots$

NB: $v_s = 0; \forall h = \text{even}$

simulation study

<https://www.multisim.com/content/tEpSbV89Gwfk3NGiU2L7J/full-bridge-inverter-km/open/>

comparative study between half and full bridge

- average output voltage:
 - ▶ Full bridge : V_s
 - ▶ Half bridge: $\frac{V_s}{2}$
- rms of the fundamental voltage V_{01}
 - ▶ Full bridge : $\frac{4V_s}{\sqrt{2}\pi} = 0.90V_s$
 - ▶ Half bridge: $\frac{2V_s}{\sqrt{2}\pi} = 0.45V_s$
- instantaneous voltage
 - ▶ Full bridge : $v_0 = \sum_{h=1}^{\infty} \frac{4V_s}{h\pi} \sin h\omega t$
 - ▶ Half bridge: $v_0 = \sum_{h=1}^{\infty} \frac{2V_s}{h\pi} \sin h\omega t$
- THD : 48.34% in both cases

References



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